



Cognitive abilities and portfolio choice

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ABSTRACT

We study the relation between cognitive abilities and stockholding using the recent Survey of Health, Ageing and Retirement in Europe (SHARE), which has detailed data on wealth and portfolio composition of individuals aged 50+ in 11 European countries and three indicators of cognitive abilities: mathematical, verbal fluency, and recall skills. We find that the propensity to invest in stocks is strongly associated with cognitive abilities, for both direct stock market participation and indirect participation through mutual funds and retirement accounts. Since the decision to invest in less information-intensive assets (such as bonds) is less strongly related to cognitive abilities, we conclude that the association between cognitive abilities and stockholding is driven by information constraints, rather than by features of preferences or psychological traits.

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1. Introduction

Transaction and information costs, broadly interpreted, have been offered as the main reason why many individuals do not invest in either stocks or other financial assets (Haliassos and Bertaut, 1995; Vissing-Jorgensen, 2004). The exact nature of these costs, however, is not well understood, and isolating the factors that prevent large sectors of the population from holding stocks is a challenge for research in household finance. In this paper we focus on the lack of cognitive abilities (as measured by mathematical and verbal skills and memory functioning) as a potential explanation for limited financial market participation.

There are several channels through which cognitive skills might affect the decision to buy stocks and other financial assets. First, the cost of gathering and processing information is lower for skilled individuals; therefore, low skills can act as a barrier preventing stockholding. Second, cognitive skills tend to be associated with preference features, such as risk aversion, that have a major influence on the willingness to bear financial risk. Third, the perception of risk is also likely to depend on cognitive abilities: low cognitive skills can make some investors overestimate the precision of the information that they possess. Overconfident investors trade more and take more financial risk than rational agents with unbiased perceptions, which implies a negative relation between cognitive skills and stockholding.

In order to study the impact of cognitive factors on stockownership, we exploit within and across country variability in cognitive abilities, stockholding and socioeconomic variables found in data from the first wave of the Survey of Health,

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Ageing and Retirement in Europe (SHARE), which surveys people aged 50 and above in 11 European countries: Austria, Belgium, Denmark, France, Germany, Greece, Italy, the Netherlands, Spain, Sweden, and Switzerland. The survey asks detailed questions on demographics, physical and mental health, employment, income, assets, social activities, and expectations. All questions are standardized across countries, thus allowing consistent international comparisons.

Most importantly for our purposes, SHARE provides detailed information on reading, mathematical, and recall ability. Cognitive abilities are known to vary across countries, even after taking education into account. For instance, evidence from the OECD Programme for International Student Assessment (PISA) shows that among 15 years olds with the same level of schooling there is large international variability in mathematical and science test scores, and that there is a consistent North–South gradient in skills (Hanushek and Woessmann, 2009). Given the age composition of our sample, we need to recognize in our empirical analysis that the elderly face a substantial mortality risk (especially at advanced ages), which reduces their planning horizon. Moreover, while the retired face more limited labor income risk, they also face much higher uncertainty about medical expenditures. As a result, it is particularly important in our context to control for the effects of investors' horizon, health status and bequest motives on the willingness to bear financial risk.

To the best of our knowledge, our paper is the first one to empirically demonstrate the effect of cognition on stockholding. We indeed find that cognitive impairment, as manifested by a reduced ability to perform numerical calculations, to read and to recall, has a strong negative influence on stock market participation. Furthermore, this result is net of the effects of age, education, health and economic resources, and it is also robust for different specifications of the cognition and other explanatory variables.

Given that the willingness to bear financial risk may change over the life-cycle, in our analysis we need to disentangle the effect of cognition on stockholding from the corresponding effects of age. We therefore split our sample into age groups (younger and older than 65), which allows us to check whether the correlation between cognitive abilities and stockholding is mainly driven by the choice of the oldest segment of the population.³ Our results show that there are no appreciable differences between age groups, which implies that ageing is not the main reason behind the correlation between cognition and stock market participation. On the other hand, we find that the propensity to hold bonds, which require less information than stocks, is affected by cognitive abilities very weakly or not at all; therefore, the correlation between cognition and stock market participation is likely due to informational barriers.

Finally, our paper offers a systematic account of the main factors affecting household portfolio choice in Europe using fully comparable microeconomic data. It represents therefore a significant improvement over previous work based on individual country data.

Understanding the reasons for financial market participation is important for many reasons. Cocco et al. (2005) show that in calibrated life-cycle models the welfare loss from no stockholding is between 1.5 and 2 percent of consumption. Limited participation and changes in participation over time are also relevant for the equity premium puzzle (Mankiw and Zeldes, 1991; Attanasio et al., 2002), the distribution of wealth (Guvenen, 2006), ownership of individual retirement accounts (Bernheim and Garrett, 2003), and wealth effects on consumption (Dynan and Maki, 2001).

From a policy point of view, European pension reforms are likely to increase reliance on individual retirement accounts and thus investors' exposure to stock market investment. These investment opportunities could bring higher expected returns, but excessive or ill-advised trading of stocks can significantly reduce realized returns. In addition, poor judgment in allocating retirement wealth can create major financial distress at a point in the life-cycle where the potential for offsetting adjustments is quite limited. Therefore, policies that aim to improve the quality of investors' financial information and awareness depend crucially on the influence of cognitive abilities on financial decisions.

The rest of the paper is organized as follows. Section 2 provides a simple framework to understand how cognition might affect stock market participation. Section 3 describes the microeconomic data and our three indicators of cognitive abilities, and Section 4, the set of variables that will be used in the empirical analysis to explain stockholding decisions. Section 5 presents the empirical results for the probability of investing in stocks, either directly or through mutual funds and investment accounts, performs robustness analysis and compares the results with the determinants of bondholding. The results are summarized in Section 6.

2. The effect of cognitive abilities on stock market participation

There are at least three different channels through which cognitive abilities might affect the decision to invest in stocks. First of all, managing a portfolio requires a specific human capital investment, in terms of time and effort needed to familiarize oneself with the notions of transaction costs, asset returns, volatility, and covariance between assets returns. Information costs represent therefore a significant barrier to entry in the stock market. Low cognitive abilities are likely to raise such costs. Campbell (2006) argues that information barriers to invest in stocks come from psychological factors that make stockownership uncomfortable for some households. Korniotis and Kumar (2006) point out that the perception of having limited abilities might also increase the cost of stock market participation.

³ We further control for many other variables that vary with age, such as health status, economic resources, bequest expectations, depression, religious participation, and social activities.

A second channel between the participation decision and stockholding is the potential relation between cognitive abilities and the curvature of the utility function. Indeed, starting with [Kahneman and Tversky \(1979\)](#), a growing strand of literature challenges the assumptions of exponential discounting and expected utility maximization.⁴ [Benartzi and Thaler \(1995\)](#) and [Barberis et al. \(2006\)](#) cite myopic loss aversion and narrow framing as potential explanations for limited participation in financial markets. [Frederick \(2005\)](#) reports experimental evidence relating cognitive ability, time and risk preferences. [Benjamin et al. \(2006\)](#) provide experimental evidence that greater cognitive ability is associated with more patient and less risk averse behavior.⁵ Lower cognitive skills would then be associated with a higher risk aversion coefficient, making participation less likely. [Dohmen et al. \(2007\)](#) find that in a random sample of 1000 German adults, lower cognitive ability is associated with significantly more impatient behavior, and with greater risk aversion, even controlling for educational attainment, income, and measures of credit constraints.

Finally, evidence from psychology shows that poor cognitive skills are associated with a low ability to process information and that memory affects the ability to perceive conditional probabilities and to distinguish between relevant and irrelevant information ([Spaniol and Bayen, 2005](#)). One way to formalize this concept is to posit that poor cognitive skills are associated with overconfidence ([Barber and Odean, 2001](#)). Overconfident investors underestimate the risk of stockownership, which makes them more likely to hold stocks. Experimental evidence shows that overconfidence differs considerably across individuals and tasks, and that overconfidence is more likely to manifest itself when individuals face relatively difficult tasks, such as finance ([Yates, 1990](#)). Individuals with low cognitive abilities might therefore underestimate financial risks.⁶

Our paper is also related to the recent literature on financial education, financial literacy, and portfolio choice. [Bernheim and Garrett \(2003\)](#) show that workers' saving choices are affected by the employer-based programs of financial education. [Lusardi and Mitchell \(2006\)](#) find a negative association between planning for retirement and financial education and [Graham et al. \(2005\)](#) show that investors who claim to understand investment products hold more efficient portfolios. [Van Rooij et al. \(2007\)](#) find that in a Dutch household survey those who have low financial literacy are also less likely to invest in stocks. A potential problem with this literature is the endogeneity of financial literacy when used as a determinant of stockholding. In contrast, we use measures of cognitive abilities that do not suffer from such endogeneity problems, given that they refer to basic numerical, verbal and recall abilities; therefore, they are unlikely to be affected by stockownership.

3. Measuring cognitive abilities

We study the relation between cognitive abilities and the decision to hold stocks either directly or indirectly, through managed accounts or mutual funds. We use the first wave of SHARE, which is a representative sample of those aged 50 and above in 11 European countries. The interviews took place in 2004 and 2005, and wealth data refer to December of the year prior to the interview.^{7,8} The survey covers 19,548 households and 32,405 individuals, and a wide range of topics, encompassing physical health, socioeconomic status, financial transfers, and intensity of social interaction. Some of the questions refer to the household (e.g. those on housing), while others are posed to both partners in a couple; for instance, the cognitive ability indicators, an overall assessment of health status, family networks and social activities.⁹ In these cases, we combine information on both partners (a more detailed description can be found in Appendix A.1).

In this section we describe the indicators of cognitive abilities that can be constructed using SHARE and document their international variability and relation with age and education. SHARE devotes a section of the questionnaire on measuring cognitive functions. The cognitive psychology literature identifies four main domains of ability: orientation, memory, executive function, and language. These abilities depend on both genetic endowment and environmental factors, such as the education received and the home environment during childhood, and change over time (see [Richards et al., 2004](#)). SHARE provides information on all these domains, and in this paper we focus on the ones that are more relevant for financial investment: ability to perform numerical operations (which we term numeracy), planning and executive function (fluency), and memory.

⁴ For instance, [Laibson \(1997\)](#) replaces exponential discounting with hyperbolic discounting to explain the co-movement of consumption and income during the life-cycle.

⁵ [Benjamin et al. \(2006\)](#) find that in the US Longitudinal Survey of Youth mathematical skill is associated with larger asset accumulation, better knowledge of pension plans, and greater financial market participation. Here we analyze a sample of mature individuals, and control for a much wider set of variables, including financial wealth, social activities, health status, and education.

⁶ Investor rationality is not, however, incompatible with the inability to estimate correctly the risk (and return) of stocks. [Sims \(2003\)](#) argues that cognitive skills act as an additional constraint that optimizing individuals face when making their economic decisions. Namely, low cognitive skills limit the accuracy with which investors estimate stock market returns.

⁷ The SHARE data collection has been primarily funded by the European Commission through the 5th framework program (project QLK6-CT-2001-00360 in the thematic program Quality of Life). Additional funding came from the US National Institute on Ageing (U01 AG09740-13S2, P01 AG005842, P01 AG08291, P30 AG12815, Y1-AG-4553-01, and OGHA 04-064). Data collection in Austria (through the Austrian Science Foundation, FWF), Belgium (through the Belgian Science Policy Administration) and Switzerland (through BBW/OFES/UFES) was nationally funded. The SHARE data set is introduced in [Börsch-Supan et al. \(2005\)](#).

⁸ In an earlier working paper ([Christelis et al., 2006](#)) we used data from an earlier release of SHARE data, thus obtaining results slightly different from the ones presented here.

⁹ The questionnaire and the sample design are patterned after the US Health and Retirement Survey (HRS) and the English Longitudinal Study of Ageing (ELSA). [Börsch-Supan et al. \(2005\)](#) report details on sampling, response rates, and definitions of variables.

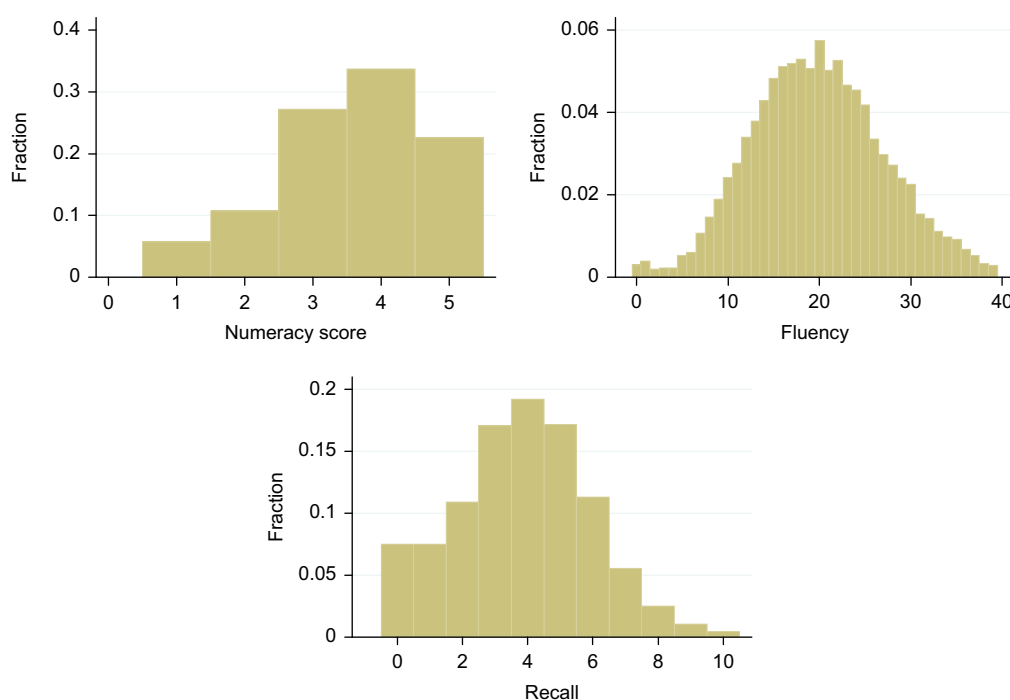


Fig. 1. Sample distributions of the indicators of cognitive abilities. *Note:* Numeracy is constructed on the basis of the number of correct answers to four basic mathematical tests. Fluency is measured as the number of animals that the interviewee can name in 1 min. Recall is measured as the number of items that the interviewee can recall in a list of 10. Variables are measured at the household level.

The indicator of numeracy measures the ability to perform basic numerical operations. This ability, in turn, affects how people make financial decisions and manage their portfolios in a manner that reflects their preferences. SHARE respondents are asked to perform the following simple calculations: (1) find 10 percent of a number; (2) find one half of a number; (3) find the number of which another known number represents two-thirds; (4) find 10 percent of another number at the end of 2 years. On the basis of these four questions Dewey and Prince (2005) construct a numeracy indicator, which ranges from one to five; details on its construction and the questions themselves are given in Appendix A.2. In Fig. 1 we report the sample distribution of numeracy (upper-left graph). The sample average is 3.4, and the standard deviation is 1.2. Less than 25 percent of the sample achieves the maximum score (5). Descriptive analysis shows that numeracy is positively associated with education and income, while it is negatively associated with age and self-perceived health.

Executive function is measured by asking respondents to name as many animals as they can in exactly 1 min. Respondents are then assigned a score, which is equal to the number of animals they can name. We call this indicator “verbal fluency”, or simply fluency. Fluency affects the ability to read and understand written texts (e.g. financial newspapers), and therefore can be an important determinant of portfolio choice. The distribution of the fluency scores is plotted in Fig. 1 (upper-right graph). The sample average is 19.4, and the standard deviation is 8.3. The bulk of the distribution (50 percent of the sample) lies between 14 and 25, but a significant number of respondents (10 percent) has a score above 32 or below 10.

In order to construct the memory indicator, respondents are first given a list of ten words, and are then asked to list which ones they remember. The indicator is constructed by counting the number of words recalled, and it ranges from zero if no word is recalled, to ten. In the rest of the paper we will refer to this indicator as recall, borrowing the terminology from the cognitive psychology literature. Recall is at the heart of the ability of comparing facts and situations at distant points in time, and accordingly can play an important role in shaping individual financial decisions. The sample mean of the distribution of the memory indicator is 3.6, and the standard deviation is 2.1. Fig. 1 (lower graph) shows that 50 percent of the samples scores between 2 and 5, and only 10 percent above 6. The SHARE data also indicate a moderate North–South gradient in cognitive abilities: basic mathematical skills, fluency, and memory are higher in Northern than in Southern Europe.

Cognitive psychologists and epidemiologists recognize age and education as important determinants of cognition.¹⁰ Therefore, it is useful to check the extent to which cognitive functions are related to age and education and if, after controlling for such observables, there is any variation left. This check is crucial since both age and education might have a

¹⁰ Schaie (1989) points out that cognitive decline starts at the age of 50 and that memory is the first function to be affected. Kaplan et al. (2001) study the effect of childhood socioeconomic environment on cognitive function in middle age.

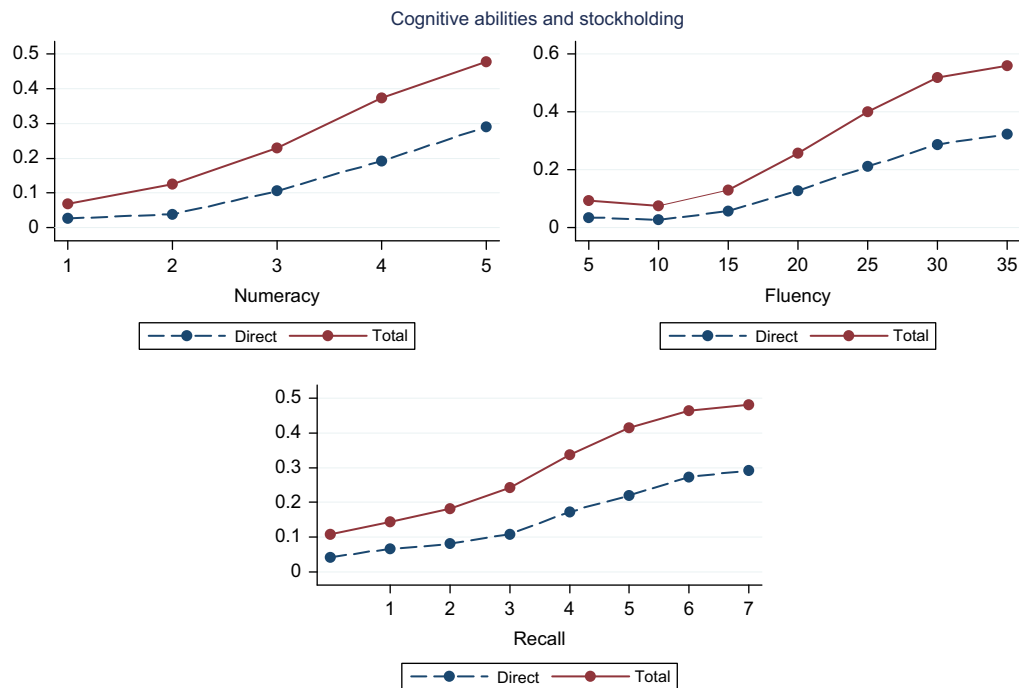


Fig. 2. Stockholding and cognitive abilities. *Note:* The figure plots the distribution of direct and total stockholding across different values of the indicators of cognitive ability. Variables are measured at the household level.

direct effect on portfolio choice, independent of their effect on cognitive functions. Descriptive analysis suggests that cognitive functions decline with age. We find, however, ample variation in cognitive abilities even among individuals belonging to the same age group. For instance, the coefficient of variation of numeracy is 0.26 in the age group 50–59 and 0.48 for those aged 80 and above. For the same age groups, the coefficients of variation of fluency are 0.35 and 0.54, respectively; for recall, 0.43 and 0.98, respectively.

Our analysis of the relation between cognitive functions and education shows that each of the three indicators is positively associated with education. As with age, there is, however, ample variation within each education group: among college graduates,¹¹ the coefficient of variation of numeracy is 0.20, and the coefficient of variation of fluency and recall are 0.32 and 0.40,¹² respectively.

Fig. 2 plots the relation between cognitive abilities and stockholding (direct and total). The figure shows that each of the indicators is positively associated with stockholding. For instance, the probability to hold stocks directly increases from 2.5 to 15.3 percent when numeracy increases from 2 to 4, while total stockholding – including also equity mutual funds and investment accounts – increases from 10.2 to 33.3 percent; the two other indicators have similar large and positive effects. The figure also shows that the association between the probability of stockholding and the cognitive indicators is approximately linear. In the rest of the paper we control for many other potential determinants of stockholding, and establish that the association between cognitive abilities and stockholding is a genuine fact, and not the result of spurious correlations. We also study which indicator of cognitive abilities is more strongly correlated with stockholding.

4. Empirical specification

While traditional finance theory predicts that investor's willingness to take financial risks depends only on risk aversion and investment opportunities, dynamic models of portfolio choice emphasize that investment opportunities and wealth itself change over time, that investors usually face other background risks, and that transaction costs, information costs and borrowing constraints limit household financial decisions. [Gollier \(2002\)](#) surveys some of the recent developments. On the empirical front, the literature has sought to single out the variables that are able to explain the patterns of portfolio choice in microeconomic data. In this section we describe how we use SHARE data to study the decision to buy stocks and other financial assets.

¹¹ This includes college degrees as well as degrees granted by vocational schools and technical institutes.

¹² The variability of cognitive skills among people with similar level of education is not a unique feature of our data. For instance, the OECD PISA shows a North–South gradient in mathematical skills among young (less than 15 years old) Europeans with the same level of schooling. [Frederick \(2005\)](#) also reports considerable differences in cognitive abilities among US students.

Table 1

Demographic and economic characteristics.

Item	Sweden	Denmark	Netherlands	Germany	Belgium	France
Direct stockholding (%)	38.8	33.2	12.8	17.1	17.7	15.6
Total stockholding (%)	71.1	56.3	24.2	24.0	38.0	42.9
Age	66.2	64.7	65.2	64.5	65.7	65.9
Couple (%)	58.9	57.5	60.2	62.7	67.3	61.5
Post-secondary education (%)	27.7	41.5	32.1	26.4	32.5	23.4
Self-employed (%)	12.1	11.6	9.9	8.6	6.7	5.7
Not working (%)	46.3	47.3	54.0	52.6	62.8	60.5
Self-reported health fair or poor (%)	18.8	35.5	50.8	38.5	39.0	45.2
Felt depressed last month (%)	47.7	45.2	53.3	46.5	53.0	61.8
Numeracy	3.8	3.7	3.8	3.8	3.6	3.3
Fluency	24.0	22.9	21.4	20.9	21.2	20.8
Recall	4.2	4.5	4.0	4.2	3.7	3.4
Social activities	0.63	0.74	0.69	0.66	0.72	0.60
Religious participation (%)	17.3	12.8	30.3	24.3	17.5	20.5
Probability of leaving an inheritance above 50,000 euro (%)	58.4	62.5	51.6	50.1	68.8	58.8
Gross household income ('000s euro)	31.6	29.2	26.8	32.7	22.8	26.0
Gross financial assets ('000s euro)	13.7	17.1	15.3	15.9	14.8	11.5
Real assets ('000s euro)	64.8	106.2	60.4	85.2	157.2	154.5
Number of households	2140	1176	2002	1954	2532	2110
Item	Switzerland	Austria	Italy	Spain	Greece	Total
Direct stockholding (%)	26.2	5.5	4.5	4.2	5.7	12.1
Total stockholding (%)	35.4	9.7	10.2	11.6	10.5	25.4
Age	64.3	65.4	65.6	66.2	65.1	65.5
Couple (%)	65.0	52.8	62.4	62.4	59.7	61.3
Post-secondary education (%)	33.9	28.0	9.6	12.6	16.9	22.9
Self-employed (%)	18.9	7.5	13.4	11.8	10.1	10.1
Not working (%)	44.7	66.7	65.1	59.1	61.9	58.2
Self-reported health fair or poor (%)	23.2	38.2	54.8	56.0	40.8	47.6
Felt depressed last month (%)	54.5	41.3	58.0	56.5	44.6	54.9
Numeracy	4.0	3.8	3.1	2.7	3.5	3.4
Fluency	21.7	23.5	15.5	16.1	15.2	19.4
Recall	4.6	3.9	3.0	2.7	3.6	3.6
Social activities	0.82	0.48	0.47	0.45	0.33	0.58
Religious participation (%)	32.3	59.9	29.9	55.1	77.7	32.6
Probability of leaving an inheritance above 50,000 euro (%)	63.1	51.1	62.7	50.5	61.8	56.4
Gross household income ('000s euro)	36.8	25.2	21.0	14.2	14.0	23.5
Gross financial assets ('000s euro)	34.3	8.5	5.1	2.7	2.9	9.8
Real assets ('000s euro)	118.9	88.3	139.4	141.6	105.9	119.4
Number of households	712	1409	1778	1753	1982	19,548

Note: Age, self-reported health, numeracy, fluency, recall, and social activities are expressed in weighted means, gross household income, gross financial, and real assets in weighted medians (adjusted for PPP). Gross financial assets are net of any stock holdings, and gross household income is net of any dividend and interest income. All variables refer to households.

4.1. Financial wealth, stockholding, and sample statistics

SHARE provides detailed information on both financial and real assets. Financial assets include bank and other transaction accounts, government and corporate bonds, stocks, mutual funds, individual retirement accounts, contractual savings for housing, and life insurance policies. The questions on real assets refer to the value of the house of residence, other real estate, business wealth and vehicles (see Christelis et al., 2005 for details).

Selected sample statistics by country are reported in Table 1. The average age of the household head is approximately 64–66 years in all countries, while the percentage of couples ranges from 53 percent in Austria to 67 percent in Belgium. Individuals in Denmark, Belgium, the Netherlands, and Switzerland are much more likely to have a post-secondary education degree. The share of households where the head (for singles) or both partners (for couples) are not working, either because they are unemployed or because they are retired, ranges between 45 percent in Switzerland to 67 percent in Austria. Household gross financial wealth also varies widely across Europe, with Switzerland clearly above the rest, followed by Sweden, while households in Italy, Spain and Greece report lower gross financial assets. The ranking between Scandinavian and Mediterranean countries is reversed if one looks at real assets, with median values of around 157,000 euro in Belgium, 139,000 euro in Italy, and 65,000 euro in Sweden.

We adopt two definitions of stockownership: direct stockownership and total stockownership, defined as stocks held directly plus stocks held through mutual funds and investment accounts (assuming that whoever holds mutual funds and

retirement accounts has some stocks in them).¹³ Table 1 reports participation in direct and total stockholding in the 11 countries of our sample. The prevalence of direct stockholding ranges from less than 5 percent in Italy and Spain to 39 percent in Sweden. Total stockholding goes from about 10 percent in Austria, Greece, and Italy to 71 percent in Sweden.¹⁴ In broad terms there is a negative gradient going from Northern to Southern Europe, with a group in the middle consisting of France, Germany, Belgium, the Netherlands, and Switzerland. Sweden and Denmark have by the far the highest prevalence of both direct and total stockholding, while Austria, Spain, Greece, and Italy are at the other end of the spectrum. The histogram suggests that country effects are potentially quite important in explaining stockholding decisions of European investors. Our regression framework pays therefore particular attention to cross-country interaction effects.

4.2. Investors' horizon

Standard finance theory suggests (assuming constant relative risk aversion preferences, no background risk, and frictionless markets) that portfolio allocations are independent of investors' horizons (Merton, 1969). If one relaxes these assumptions, then investors' age can affect portfolio allocations, as Ameriks and Zeldes (2004) point out in their exhaustive survey of this issue. The direction of the effect, however, is far from simple; it depends on the choice of the utility function, the correlation of income and rate of return shocks, transaction costs, the existence and projected trajectory of human capital and other non-traded assets, the behavior of risk aversion over the lifetime and taxes.¹⁵ Despite the ambiguous theoretical relation between age and stockholding, Ameriks and Zeldes present ample evidence that professionals and mutual fund companies recommend reducing stock exposure as one ages. For instance, a typical advice is that the asset share invested in stocks should be 100 minus investor's age. Therefore, even if there are no compelling theoretical reasons to reduce risk exposure with age, people might nevertheless do so following standard financial advice.

In our context, it is quite important to controlling for age and for the correlation between age and cognitive abilities. We introduce an age polynomial in all regressions, and further split our sample into two age groups (younger and older than 65), which allows us to check whether the correlation between cognitive abilities and stockholding changes for the oldest segment of the population.

4.3. Health status

While the elderly are unlikely to face significant income risk, except for the inflation risk associated with annuities, they typically face a much higher health risk; therefore, it is essential to take into account health in our sample.

There are several channels through which health risk can negatively affect the decision to hold stocks. Edwards (2008) shows that uninsurable morbidity risk induces investors to become effectively more risk averse, thereby reducing the asset share allocated to stocks. He documents this correlation using the Health and Retirement Study (HRS) and the Survey on Assets and Health Dynamics Among the Oldest Old (AHEAD), and finds that current health status and expectations about future health affect negatively the share of risky assets.

Qiu (2004) and Goldman and Maestas (2005) argue that health risks increase the variability of future medical expenditures. As with other background risks, health risks cannot be easily avoided or diversified away, and thus investors will tend to reduce exposure to avoidable risks, such as financial risk. This tendency leads to a negative correlation between health risk and stockownership. These studies measure health risk by whether individuals have supplemental insurance, and show that the effect of the reduced health risk through insurance is both statistically and economically significant.

A further reason why health affects stockownership is that health status is correlated with individual resources, along the lines suggested by Smith (1999). Poor health is thus associated with low wealth and, consequently, with a smaller investment in stocks. Rosen and Wu (2004) use the HRS to estimate the relation between health status, asset ownership and asset share invested in risky assets, and find that poor health tends to be negatively related to ownership and investment in risky assets. They also show that this negative association is not due to the omission of variables that simultaneously affect health and financial decisions, such as planning horizon, risk aversion, and health insurance status.

While objective health data are not affected by different reporting styles across individuals and are therefore more reliable, subjective data describe individual perceptions.¹⁶ In our context, self-reported health status might be even more relevant than objective health indicators. If people buffer health risks by investing less in stocks, then it is the perception of

¹³ The appendix reports further details on the calculation of our measures of wealth.

¹⁴ See Engstrom and Westerberg (2003) for an analysis of stock market participation in Sweden.

¹⁵ Empirically, Ameriks and Zeldes (2004) show that investors make very few changes in either the extensive or the intensive margin. They also find that investors' active choices imply, if anything, a slightly increased exposure to stocks over the life-cycle, and argue that any observed decline in stockholding with age is likely to be a cohort rather than an age effect. Given the cross-sectional nature of our data, however, in our analysis we cannot distinguish age from cohort effects in stockholding.

¹⁶ Empirically, health status is an intrinsically unobservable variable. Researchers generally seek to measure it using household surveys or medical-administrative records. Currie and Madrian (1999) suggest eight different categories of health indicators: (1) subjective health status; (2) health limitations to the ability to work or to carry normal activities; (3) functional limitations on normal activities; (4) chronic disease; (5) permanent disability; (6) health care utilization; (7) nutritional status, as measured by the body mass index; and (8) expected mortality. SHARE has information on all these measures, which allows one to check whether these measures are mutually consistent and whether health outcomes can be forecasted from subjective evaluation of health status (Börsch-Supan et al., 2005).

such risks (not necessarily their actual presence) that determines financial decisions. Furthermore, using self-reported health status delivers results that are more easily comparable to other studies using the same variable, such as Rosen and Wu (2004).

Self-reported health status is ranked on a 1–5 scale (1 = *Excellent*, 2 = *Very Good*, 3 = *Good*, 4 = *Fair*, 5 = *Poor*). A high value of the index corresponds therefore to poorer health. Self-assessed health measures are well known to correlate strongly with objective indicators (Currie and Madrian, 1999), and may directly affect economic behavior. The distribution of self-reported health status indicates that 3.1 percent of households view their health as excellent, 11.3 percent as very good, 38 percent as good, 34.3 percent as fair, and the rest as poor.¹⁷ Table 1 shows the percentage of households heads who answer that their health is fair or poor, and we observe interesting differences across countries: in Spain and Italy households claim more often that they are in poor or very poor health, while they are least likely to report so in Switzerland.¹⁸

4.4. Social activities

In addition to the information that investors can collect from media and financial advisors, there are information spillovers from informed to uninformed consumers in the same social circle. Individuals often learn about investment opportunities from others, and how this occurs depends on the specific process of social learning and on how people interact. Previous literature shows that social interaction affects portfolio outcomes.

Duflo and Saez (2002) show that in the US the decision to participate in tax deferred retirement accounts is significantly affected by a similar decision of employees in the same department. As a result, the experience of peers about the performance of their investments is passed on to others. Hong et al. (2004) show that stock market participation is higher among more socially connected individuals. Furthermore, this effect is stronger among individuals living in communities with a higher participation rate to begin with, implying that social learning interacts positively with learning induced by market development.

A related line of research points out that trust is an important determinant of economic exchange, and financial transactions in particular. Guiso et al. (2004) find that the proportion of stockholders is higher in Italian provinces with relatively high social trust, other things being equal. People more active socially might have a stronger tendency to trust, and therefore the effect of sociability and trust are difficult to distinguish empirically.

As pointed out by Hong et al. (2004), word-of-mouth and observational learning are two channels through which social interaction might influence stock market participation. In contrast to their analysis, which focuses only on whether people interact with their neighbours or take part in religious activities, we can condition on a much larger set of variables.

Our data also allow us to test if social activities amplify or dampen the effect of cognitive ability on the participation decision. In principle, the effect is ambiguous, thus making the empirical analysis more informative and interesting. On the one hand, social learning and cognitive ability can be viewed as complements, in that the effect of cognitive ability is larger among socially connected individuals. On the other hand, lack of cognitive ability can be supplemented by social learning and therefore cognitive ability is more important among less socially active investors.

In order to construct our indicator of social activities we rely on a set of questions on various kinds of social activities that individuals have performed in the month prior to the interview. The possible activities that we consider as indicating some form of social interaction are (1) participating in a sport, social or other kind of club; (2) taking part in a political or community-related organization; and (3) participation to religious organization.¹⁹ We convert the first two activities in dummy variables and construct a household level indicator of social interaction as the sum of the two dummies. We enter religious participation as a separate variable to check if religion is associated with higher stock market participation, as in Hong et al. (2004). Table 1 indicates that household religious activity ranges from approximately 13 percent in Denmark to 78 percent in Greece.

The distribution of the indicator of social activities across countries shows that in Denmark and Switzerland the index are 0.74 and 0.82, respectively, meaning that households in these countries have engaged, on average, in approximately 0.78 social activities in the previous month. The levels of social interaction in Spain and Greece are lower, averaging, respectively, to only 0.45 and 0.33 activities per household.

When looking at some of the different components of social interaction, Hank and Erlinghagen (2005) find that volunteering displays a strong North–South gradient, just as our constructed index of social activities. In addition, volunteering is positively correlated with participation in organizations and with the provision of informal help and care to other people. They also document that the latter two activities are also more prevalent in Northern than Southern countries. This finding is quite surprising, given that one might expect that providing help would be more prevalent in the South, at least within families.

¹⁷ Individuals in SHARE are also asked if they suffer from chronic disease and disability. Chronic diseases affect over one-third of the sample, and 13 percent report some form of disability (such as blindness, deafness, or disabilities limiting mobility).

¹⁸ It is worth noting that some of these differences may be due to differences in reporting styles across countries that result in different evaluations of similar physical problems (Jürges, 2005a).

¹⁹ SHARE also has information on voluntary work, care for a sick adult, helping family and friends, attendance of an educational or training course. The coefficients of these indicators were not statistically different from zero and are therefore excluded from the final specification.

4.5. Bequest motives

One reason why portfolios of the elderly might differ from those of other investors is that the elderly face mortality risk, and have, as a result, shorter horizons than younger people. The time horizon for a couple as a unit is even shorter (Hurd, 2002). While both young and old persons might have a bequest motive, Hurd (2002, p. 433) argues that “... for a young person the event of a bequest is so remote as not to alter behavior. For the elderly, however, a bequest motive could extend the time horizon, reducing or eliminating any effects of mortality risk”. For this reason, we control for bequest motives and intention to leave a bequest in our sample of elderly individuals. Our expectation is that the intention to leave a bequest impacts portfolio allocations in the same direction as investors’ horizons.

SHARE contains three questions on the expectation to leave an inheritance, which are asked to all financial respondents in the household. The first question asked is whether there is a chance that the person interviewed expects to leave an inheritance greater than 50,000 euro. If the answer is affirmative, then the next question asked is whether respondents expect to leave an inheritance greater than 150,000 euro. If, on the other hand, the answer to the first question is negative, then they are asked whether there is a chance that they will leave any inheritance. We use the first answer to construct an indicator of the probability to leave an inheritance (details on the construction and the original questions can be found in Appendix A.4).

Table 1 reports the distribution of the expected probability of leaving an inheritance across countries, which ranges from 69 percent in Belgium to 50 percent in Germany and Spain. The expectation of leaving an inheritance is strongly positively correlated with current wealth, as shown by Jürges (2005b), who also finds that roughly one-third of the households expect to bequeath at least half of their current wealth.

5. Econometric results

To model the participation decision, households compare the utility gain from owning stocks with the entry cost. We express the net utility gain as

$$y_h^* = z_h' \delta + \varepsilon_h \quad (1)$$

where h is the household index, and z_h are the observable variables affecting the utility gain from owning stocks. Utility also depends on unobservable variables, which enter (1) through ε_h , a standard normally distributed random term. Household h owns stocks if $y_h^* \geq 0$, which implies that the probability of observing stockownership is $\Phi(z_h' \delta)$. If the unobserved factors ε_h are normally distributed, then we can use the probit model and estimate δ with standard maximum likelihood methods.

In our estimation we also carefully address item non-response, which can bias estimates and also lead to a substantial efficiency loss due to the large number of variables included in the regression. SHARE data are imputed using a multiple imputation methodology, with five implicate datasets that contain the imputed values generated from the original dataset.²⁰ Given that the data are multiply imputed, we estimate separately our model on each of the five implicate datasets and then combine the coefficients and the standard errors using the results in Rubin (1987).²¹ Coefficients are computed as the means of the coefficients obtained from the five implicates, while standard errors account for within and between imputation variability of the five estimates. Appendix A.1 and Christelis (2008) provide more details on the imputation procedure used.

5.1. Baseline results

In order to single out the determinants of financial market participation, we study separately direct and total stockholding, which also includes stocks owned through managed investment accounts and mutual funds. We obtain similar results when examining indirect participation only; therefore, for brevity, we do not report results from these regressions.

We assume that stock market participation depends on demographic variables such as a dummy for couples, education, a dummy for not working, which is meant to account for the possible effect of public pensions on stock market participation, indicators of household resources (gross financial and real assets and household disposable income), a self-reported fair or poor health status dummy, a dummy for self-employment that serves as a rough proxy for risk aversion, a second-order age polynomial, indicators of religious and social activities and expectations to leave an inheritance. To proxy for the role of overconfidence we add to the regressions also a dummy variable indicating depression. Finally, stock market participation is related to the three cognitive function indicators.

²⁰ This is also the number of implicates used in the US Survey of Consumer Finances (Kennickell, 1991). The bias of the standard error of the estimates depends on the number of implicates and on the fraction of missing data. Rubin (1987) shows that this bias decreases with the number of implicates.

²¹ If one adopts the most stringent definition of imputed values, i.e. a value of a household-level variable (like direct or total stockholding) is considered imputed if any one of its constituent parts is imputed for any household member, then the proportions of imputed values for direct and total stockholding are 6.2 percent and 7.7 percent. For wealth, there are 28.8 percent of households in which at least one component of real wealth values is imputed (55.7 percent in case of financial wealth).

Table 2
The effect of cognitive abilities on stockholding.

Variable	Direct participation		Total participation	
	With education (1)	No education (2)	With education (3)	No education (4)
Age	0.000 (0.000)	0.000 (0.000)	−0.001 (0.000) **	−0.001 (0.000) **
Numeracy	0.015 (0.003) **	0.018 (0.003) **	0.015 (0.004) **	0.019 (0.003) **
Fluency	0.001 (0.000) *	0.001 (0.000) **	0.002 (0.001) *	0.002 (0.001) *
Recall	0.005 (0.001) **	0.006 (0.001) **	0.006 (0.002) **	0.006 (0.002) **
Self-reported health fair or poor	−0.019 (0.005) **	−0.022 (0.005) **	−0.014 (0.006) *	−0.016 (0.006) **
Social activities	0.010 (0.004) **	0.011 (0.004) **	0.011 (0.005) *	0.013 (0.005) *
Probability of leaving an inheritance	0.043 (0.007) **	0.045 (0.007) **	0.054 (0.008) **	0.055 (0.008) **
Post-secondary education	0.059 (0.008) **		0.051 (0.009) **	
Household income (in 1000 euro)	0.0005 (0.000) **	0.001 (0.000) **	0.0009 (0.000) **	0.001 (0.000) **
Financial assets (in 1000 euro)	0.003 (0.000) **	0.000 (0.000) **	0.010 (0.000) **	0.000 (0.000) **
R ²	0.24	0.24	0.47	0.46
Observations	19,548	19,548	19,548	19,548

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, age, age squared, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a dummy for participation in social activities, a full set of country dummies, and interaction of country dummies with financial wealth. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars, at the 1 percent level. The R² measure is computed as in Estrella (1998).

Given the different institutions and constraints facing investors in each country, we start out with a flexible specification in which we interact each variable with the country dummies.²² This means that the effect of, say, numeracy, social activities or health status in the initial specification is not restricted to be the same in the 11 countries of the sample. For each regression, we then test the hypothesis that the coefficients of all variables, except for education, financial wealth and the country dummies, are the same across countries. The hypothesis is not rejected at standard confidence levels (*p*-values were equal to 0.77 for direct stockholding and to 0.17 for total stockholding).²³ We therefore present the estimates for the restricted specification; the results for the unrestricted models are available upon request.

We report the estimated average of the marginal effects evaluated at each observation and the associated standard errors of the main variables of interest. The average of the individual marginal effects measures the average impact on the population of a change in the variables. This measure is preferable to the commonly used practice of evaluating the marginal effects at the mean of the regressors, given that the latter can lead to seriously biased results (Train, 2003). The estimated marginal effects are averages of nonlinear functions of the coefficients and thus the computation of their standard errors is not straightforward. We calculate these standard errors by using 200 bootstrap replications. All probit coefficients and the associated standard errors from the two baseline regressions for direct and total participation are reported in Table A1 (the coefficients of the other regressions are available upon request).

Results from the baseline specification are reported in Table 2, where columns (1) and (2) refer to direct and (3) and (4) to total stockholding. In order to save space and increase readability, we report only a subset of the marginal effects. Our estimates imply that the effect of the cognitive indicators is positive, precisely estimated and economically important. If numeracy increases by 1.2 units (equivalent to one standard deviation of the variable), the probability of direct and total stockownership increases by $1.5 \times 1.2 = 1.8$ percentage points. Since the sample average of direct stockholding is 12 percent, an increase in numeracy of one unit is associated with an increase of stockholding of about 15 percent of its initial value. In the case of total stockholding, the effect of an increase in numeracy by one standard deviation would be also equal to 1.8 percentage points, which is 7 percent of total stockholding. Raising fluency by one standard deviation (8.3 points) increases the probability of direct stockholding by $0.001 \times 8.3 = 0.83$ percentage points and that of total stockholding by

²² Our framework does not explicitly account for cross-country differences in tax regimes and other country-specific institutions. This is one reason why all our regressions employ a full set countries dummies and why we test for the stability of the coefficients across countries.

²³ Hypothesis testing was conducted using F-tests that account for multiple imputation as in Li et al. (1991).

1.7 percentage points. Finally, raising the recall indicator by one standard deviation (2.1) is associated in direct and total stockholding with increases of 1.1 and 1.3 points, respectively. Overall, the results support the hypothesis that higher cognitive abilities raise stock market participation.²⁴

Table 2 also reports other interesting results. The effect of health status on stockholding is negative, possibly because poor health is associated with higher health risk and expenditures, higher risk aversion or lower household resources, confirming the empirical findings of Rosen and Wu (2004) from US data. Socially active households are more likely to own stocks either directly or indirectly by about 1 percentage point. Raising the chance of leaving an inheritance from zero to 1 is associated with an increase in the probability of stockholding of roughly 5 percentage points (4.3 percent for direct and 5.4 for total stockholding, as shown in columns 1 and 3 of Table 2). The marginal effect of age is negative and statistically different from zero for total stockholding: a 10-year age difference is associated with a reduction of the probability of 1 percentage point.

The marginal effect of the post-secondary education dummy reveals how much differences in education affect stock market participation. The effect is equal to 5.9 percentage points for direct stockholding and 5.1 points for total stockholding. These effects are large: they might reflect not only the fact that individuals with higher education are more financially sophisticated and thus face lower costs of stockholding, but also differences in permanent income and resources.

We use financial wealth and income as further measures of household resources. Financial wealth is strongly associated with the probability of becoming a stockholder.²⁵ The most natural interpretation is that fixed transaction costs prevent poor households from owning stocks. The probit regression indicates that raising financial wealth by 1000 euro increases the average probability of direct and total stockholding by 0.3 and 1 percentage points, while raising income by 1000 euro leads to corresponding increases of around 0.05 and 0.09 percentage points. It is worth noting that in the probit regressions financial wealth is interacted with the country dummies, and that in Table 2 we report the average effect across countries.

Given that cognitive skills are partly acquired through education, it is also interesting to see what happens when we omit education from the regression. The estimated effects in columns (2) and (4) of Table 2 represent the “gross effects” of cognitive abilities on participation. The effect of numeracy (when augmented by one standard deviation) for direct stockholding increases from 1.8 to 2.2 percentage points with respect to the specification that includes education, while for total stockholding it increases from 1.8 to 2.3 percentage points. On the other hand, omitting education leaves the effects of fluency and recall essentially unchanged.

5.2. Sample splits

As discussed in Section 4.2, Ameriks and Zeldes (2004) point out that the demand for stocks might change over the life-cycle, although there is considerable uncertainty as to the direction of this effect. Given that cognitive abilities decline over time and that the elderly are less likely to hold stocks, it is important to separate the effect of cognition from any other factors that might affect the age-profile of stockholding. In particular, one wonders if our results are driven by the oldest portion of our sample.

Therefore, we split the sample by age (younger or older than 65, which is the male retirement age in most of our countries), and re-estimate our baseline regression in both subsamples. This procedure allows us to have different effects of all our explanatory variables between the two samples, which should capture the special circumstances of those over 65, such as any need to change risk exposure. Should such need exist, it could stem from various factors that we capture with our regressors. For example, the influence of health problems could be reflected in our self-reported health variable, bequest considerations in our variable denoting the probability to leave an inheritance and economic resources in our household income and wealth variables. Most importantly, as in the baseline regression, we control for age itself (by using a 2nd-degree polynomial) in both subsamples, which should help us capture other age-dictated reasons for reducing risk, assuming that no cohort effects are present.

Our estimates, shown in Table 3, indicate that the marginal effect of age on direct stockholding is not statistically different from zero in either of the two subsamples. On the other hand, we find a negative and significant effect of age (–0.2 percentage points for one additional year) on total stockholding for those older than 65. This result indicates that those belonging to the oldest segment of the population tend to withdraw from stockholding through mutual funds and retirement accounts as they age.

Although there are some differences between the two subsamples,²⁶ the marginal effects of all three cognitive variables are broadly similar in both age groups. Therefore, these results suggest that our sample of elderly households does not bias the results towards finding a positive association between stockholding and cognitive abilities. In addition, given that the estimated effect of our cognition variables is net of the effects of all the aforementioned explanatory variables and age, it should also be net of the aged-related reasons for reducing stockholding that these variables capture.

²⁴ Computing the change in the probability of owning stocks when each of the three indicators increases from the 25th to the 75th percentile of its distribution leads to similar results.

²⁵ The measure of financial wealth used in our regressions excludes the value of all stockholdings.

²⁶ The Chow tests for equality of coefficients suggest that the 2 samples are statistically different for total stockholding (see column 3 in Table A2), but the results are less clear for direct stockholding (column 1 in Table A2).

Table 3

The effect of cognitive abilities on stockholding by age.

Variable	Direct participation		Total participation	
	Age less than 65 (1)	Age more or equal to 65 (2)	Age less than 65 (3)	Age more or equal to 65 (4)
Age	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	−0.002 (0.001) **
Numeracy	0.018 (0.005) **	0.011 (0.003) **	0.016 (0.006) **	0.015 (0.004) **
Fluency	0.001 (0.001) *	0.001 (0.000)	0.002 (0.001) *	0.001 (0.001)
Recall	0.005 (0.002) **	0.005 (0.001) **	0.004 (0.003)	0.007 (0.002) **
Self-reported health fair or poor	−0.022 (0.008) **	−0.013 (0.006) *	−0.024 (0.010) *	−0.008 (0.007)
Social activities	0.010 (0.004) **	0.010 (0.004) **	0.011 (0.005) *	0.011 (0.005) *
Probability of leaving an inheritance	0.049 (0.011) **	0.035 (0.008) **	0.070 (0.014) **	0.042 (0.009) **
Post-secondary Education	0.066 (0.011) **	0.048 (0.010) **	0.063 (0.013) **	0.045 (0.012) **
Household Income (in 1000 euro)	0.001 (0.000) **	0.000 (0.000)	0.001 (0.000) **	0.001 (0.000) **
Financial Assets (in 1000 euro)	0.004 (0.000) **	0.003 (0.000) **	0.011 (0.000) **	0.009 (0.000) **
R ²	0.24	0.22	0.47	0.41
Observations	10,545	9003	10,545	9003

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a dummy for participation in social activities, a full set of country dummies, and interaction of country dummies with financial wealth and post-secondary education. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars at the 1 percent level. The R² measure is computed as in Estrella (1998).

Hong et al. (2004) have emphasized the role of social activities as one of the channels through which people learn about financial matters, and therefore as one means of enhancing stock market participation. They show that investors who are more active socially are more likely to own stocks and that a social multiplier operates in such a way as to magnify the effect of the determinants of stockholding among more socially active individuals.

In order to establish whether such a social multiplier operates in SHARE countries as well, we split the sample between social (53.6 percent of the sample) and non-social investors (46.4 percent) and estimate the model within each of the two subsamples. The social multiplier hypothesis suggests that the effect of cognitive ability is higher in the sample of more socially active investors. The results reported in Table 4 imply that the effect of numeracy and recall on direct and total stockholding is larger among socially active households, while the effect of fluency is similar across the two subsamples and perhaps stronger for total participation in the no social activities subsample. These findings suggest that social interactions magnify the effect of mathematical skills and memory on stock market participation.²⁷

Social interactions also amplify the effect of the chance of leaving an inheritance and of education on direct and total participation, which is again consistent with the existence of a social multiplier. Among the more socially connected, the effect of post-secondary education on direct stockholding is 6.6 percentage points, against 5.2 points in the sample of individuals who do not engage in social activities. The pattern of results is similar for total participation.

5.3. Robustness checks

In this section we perform several robustness checks. In particular, we investigate how our results are affected by alternative parameterizations of the variables denoting cognition and economic resources, additional proxies for the

²⁷ In this case, a Chow test shows that the hypothesis that the coefficients of the two regressions are equal is not rejected (columns 2 and 4 in Table A2); therefore, the differences in the marginal effects of cognitive variables across the two subsamples are due to differences in the distribution of household characteristics (including cognition).

Table 4

The effect of cognitive abilities on stockholding, by participation in social activities.

Variable	Direct participation		Total participation	
	No social activities (1)	Social activities (2)	No social activities (3)	Social activities (4)
Age	0.000 (0.000)	0.001 (0.000) *	−0.001 (0.001)	−0.001 (0.001)
Numeracy	0.011 (0.003) **	0.018 (0.005) **	0.013 (0.004) **	0.017 (0.005) **
Fluency	0.001 (0.000) **	0.001 (0.001)	0.002 (0.001) *	0.001 (0.001)
Recall	0.003 (0.002) *	0.007 (0.002) **	0.003 (0.003)	0.008 (0.002) **
Self-reported health fair or poor	−0.013 (0.007)	−0.026 (0.011)	−0.008 (0.008)	−0.020 (0.011)
Probability of leaving an inheritance	0.031 (0.009) **	0.054 (0.011) **	0.039 (0.012) **	0.069 (0.011) **
Post-secondary education	0.052 (0.011) **	0.066 (0.010) **	0.049 (0.013) **	0.056 (0.012) **
Household income (in 1000 euro)	0.000 (0.000) *	0.001 (0.000) **	0.001 (0.000) **	0.001 (0.000) **
Financial assets (in 1000 euro)	0.003 (0.000) **	0.004 (0.000) **	0.009 (0.000) **	0.010 (0.000) **
R ²	0.21	0.24	0.44	0.46
Observations	9064	10,484	9064	10,484

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a full set of country dummies, and interaction of country dummies with financial wealth and post-secondary education. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each impute dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars at the 1 percent level. The R^2 measure is computed as in Estrella (1998).

bequest motive and for longevity, our imputation procedure, and the interaction between partners during the interview. The results for direct participation are reported in Table 5, and those for total participation in Table 6. For brevity, we focus only on the marginal effects of the cognition variables.

Column (1) in Tables 5 and 6 shows the results when all cognitive variables enter our specification in dummy form. The evidence confirms the positive link between cognition and stockholding (direct and total), though the marginal effects are sometimes less precisely estimated. Precision improves in column (2), where numeracy enters as a categorical variable, while recall and fluency enter linearly in the regression. Again the association between cognition and stock market participation is positive.

In order to check whether these alternative parameterizations of the effect of cognition on participation provide a better description of the data, we compute the Bayesian Information Criterion (BIC) for the baseline specification (with all cognitive variables in linear form), for the specification where all cognition variables appear as categorical variables, and for the specification where only numeracy enters as a categorical variable. The results for direct and total participation are reported in Table A3 of the appendix and show that for both direct and total participation the baseline specification has a much more negative BIC, which clearly indicates a better fit. Therefore, the test lends support to the specification in which all three cognitive variables enter linearly.

A number of households refuse to answer the cognitive abilities questions. In order to check whether there is any effect of the imputation procedure, we estimate the model using only the non-imputed observations for the cognitive variables. This check is necessary, given that imputations are based on variables that might have an independent effect on participation. Column (3) in Tables 5 and 6 reports the results, which are basically unchanged with respect to the baseline specification, for both direct and total stockholding.

We also investigate whether financial wealth and income might have a nonlinear effect on participation, perhaps due to transaction costs affecting stockholding only at low levels of income and wealth. This check is done by using dummies for financial wealth and income quartiles, a specification that might also reduce the impact of measurement error in income and wealth. The results are reported in column (4) of Tables 5 and 6, and are consistent with those of the baseline specification (Table 2). Therefore, while nonlinearities in financial wealth and income are potentially important, they do not obscure the effect of cognition on participation.

To explore the possible effect of planning horizon on participation, we use two additional indicators of bequest expectations: the chance of leaving an inheritance larger than 150,000 euro and that of leaving any positive amount. The results in column 5 of Tables 5 and 6 are similar to the baseline specification for both direct and total stock market participation. We also explore the effect of alternative indicators of planning horizon, such as life expectancy at the age of

Table 5

Robustness checks for direct participation.

Variable	All cognition variables in dummy form	Cognition in dummy form, fluency and recall linear	Without observations with imputed cognitive variables	With quartiles of income and wealth	With all three bequest expectations	Using life expectancy for those aged 65 and less	Using life expectancy for those aged more than 65 and less than 70	Partner present during the cognitive tests
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Numeracy			0.016 (0.003) **	0.014 (0.003) **	0.014 (0.003) **	0.018 (0.005) **	0.018 (0.008) *	0.019 (0.005) **
Fluency		0.001 (0.000) *	0.001 (0.000) **	0.001 (0.000) *	0.001 (0.000) *	0.001 (0.001) *	0.001 (0.001)	0.001 (0.000)
Recall		0.005 (0.001) **	0.005 (0.001) **	0.005 (0.001) **	0.005 (0.001) **	0.006 (0.002) **	0.002 (0.004)	0.008 (0.002) **
2nd level of numeracy	−0.021 (0.016)	−0.021 (0.016)						
3rd level of numeracy	0.005 (0.015)	0.005 (0.015)						
4th level of numeracy	0.020 (0.015)	0.020 (0.015)						
5th level of numeracy	0.035 (0.015) *	0.034 (0.015) *						
2nd level of fluency	0.022 (0.006) **							
3rd level of fluency	0.033 (0.008) **							
2nd level of recall	0.009 (0.007)							
3rd level of recall	0.024 (0.008) **							
R ²	0.24	0.24	0.24	0.24	0.24	0.24	0.22	0.25
Observations	19,548	19,548	14,952	19,548	19,548	9691	1592	12,521

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, age, age squared, education, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a full set of country dummies, and interaction of country dummies with financial wealth and post-secondary education. The 2nd level of fluency reflects a score higher than 15 and less or equal to 25 words, while the 3rd level a score of more than 25 words. The corresponding scores for the 2 levels of recall are 2 and 5 words. The two additional questions about expectations to leave a bequest refer to questions about any bequest and a bequest higher than 150,000 euro. The target age for the life expectancy question is 75 for those aged 65 and less, and 80 for those aged between 65 and 70. Dummies for the presence of the other partner for one case and both cases in a couple are used. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars at the 1 percent level. The R² measure is computed as in Estrella (1998).

65 and less, and life expectancy between 65 and 70.²⁸ The results are reported in column 6 of Tables 5 and 6 and are, if anything, stronger than those from the baseline specification.

The last column of Tables 5 and 6 adds a dummy for the presence of the partner during the cognitive function test. Identifying the precise channel through which the presence of one partner might affect the cognitive test score of the other partner is beyond the scope of this paper. However, given that such an effect might exist, adding a variable to control for whether the partner is present during the interview could be informative. The results are, however, very similar to those of the baseline specification, which seems to suggest that the presence of the partner does not affect the relation between participation and cognitive abilities.

We thus conclude that our findings are robust to a number of checks and point toward a positive association between stock market participation and cognitive abilities. In order to examine what lies behind this association, we turn to the next section.

²⁸ In SHARE individuals aged 65 and less are asked to give the chance of survival to the age of 75 and those aged between 65 and 70 to the age of 80.

Table 6

Robustness checks for total participation.

Variable	All cognition variables in dummy form	Cognition in dummy form. fluency and recall linear	Without observations with imputed cognitive variables	With quartiles of income and wealth	With all three bequest expectations	Using life expectancy for those aged 65 and less	Using life expectancy for those aged more than 65 and less than 70	Using indicator for presence of partner
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Numeracy			0.016 (0.003) **	0.015 (0.003) **	0.015 (0.004) **	0.019 (0.005) **	0.021 (0.011)	0.023 (0.005) **
Fluency		0.002 (0.001) *	0.002 (0.000) **	0.002 (0.001) *	0.002 (0.001) *	0.002 (0.001) *	0.003 (0.001) *	0.002 (0.001)
Recall		0.006 (0.002) **	0.006 (0.002) **	0.006 (0.002) **	0.006 (0.002) **	0.004 (0.003)	0.008 (0.006)	0.007 (0.003) *
2nd level of numeracy	0.003 (0.017)	0.004 (0.017)						
3rd level of numeracy	0.012 (0.017)	0.015 (0.018)						
4th level of numeracy	0.038 (0.018) *	0.041 (0.018) *						
5th level of numeracy	0.048 (0.019) *	0.049 (0.019) *						
2nd level of fluency	0.030 (0.008) **							
3rd level of fluency	0.046 (0.012) **							
2nd level of recall	0.025 (0.009) *							
3rd level of recall	0.030 (0.011) **							
R ²	0.47	0.47	0.47	0.46	0.47	0.48	0.36	0.48
Observations	19,548	19,548	14,952	19,548	19,548	9691	1592	12,521

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, age, age squared, education, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a full set of country dummies, and interaction of country dummies with financial wealth and post-secondary education. The 2nd level of fluency reflects a score higher than 15 and less or equal to 25 words, while the 3rd level a score of more than 25 words. The corresponding scores for the 2 levels of recall are 2 and 5 words. The two additional questions about expectations to leave a bequest refer to questions about any bequest and a bequest higher than 150,000 euro. The target age for the life expectancy question is 75 for those aged 65 and less, and 80 for those aged between 65 and 70. Dummies for the presence of the other partner for one case and both cases in a couple are used. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars at the 1 percent level. The R² measure is computed as in Estrella (1998).

5.4. Bondholding

Not all assets require the same degree of sophistication. Dealing with lower-return, lower-risk assets, such as bonds, typically requires a lower degree of financial sophistication than stocks. The degree of financial sophistication and the ability to process economic and financial data are in turn affected by cognitive abilities. Therefore, one might expect that the demand for less information-intensive assets, such as bonds, is less affected by cognitive impairment. On the other hand, if cognitive impairment prevent individuals from investing in any financial assets, then we would have to conclude that information barriers are not likely to explain the correlation between cognitive abilities and stock market participation.

In Table 7 we therefore estimate separate probit regressions for bonds, using the same specification as in Table 2. The effect of numeracy (0.5 percentage points) and recall (0.3 percentage points) is smaller than for stocks, while that of fluency is not statistically different from zero.

We also check if the results are driven by the correlation between cognitive functions and participation in particular age groups. This is done by estimating the bond market participation equation in the same subsamples as in Table 3 (younger or older than 65). The results are reported in columns (2) and (3) of Table 7 and confirm that cognitive variables have a

Table 7

The effect of cognitive abilities on bondholding.

Variable	Whole sample (1)	Age less than 65 (2)	Age more or equal to 65 (3)
Numeracy	0.005 (0.002) *	0.004 (0.004)	0.005 (0.003)
Fluency	0.001 (0.000)	0.001 (0.000) *	0.000 (0.001)
Recall	0.003 (0.001) **	0.001 (0.002)	0.005 (0.001) **
R ²	0.20	0.18	0.23
Observations	19,548	10,545	9003

Note: Mean of marginal effects evaluated at each observation. Each regression includes also real assets, a dummy for couples, age, age squared, education, a dummy for feeling depressed the previous month, dummies for being self-employed and not working, a dummy for religious participation, a dummy for participation in social activities, a full set of country dummies, and interaction of country dummies with financial wealth and post-secondary education. Standard errors are reported in parenthesis, calculated using 200 bootstrap replications. The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars, at the 1 percent level. The R^2 measure is computed as in Estrella (1998).

larger effect on stockholding than on bondholding.²⁹ Overall, the results for bondholding suggest that information barriers, rather than preference heterogeneity, can indeed explain the association between cognitive abilities and portfolio decisions.

5.5. Share of stocks in financial assets

There are a number of reasons why cognitive functioning might affect not only participation in the stock market but also the share of financial assets invested in stocks. For example, it might help people realize the superior risk-adjusted performance of stocks relative to bonds or cash in the long run, compare the returns of individual stocks as well as the features of different mutual funds. As a result, people with higher cognitive abilities might be more comfortable with the idea of investing a larger share of their portfolio in stocks.

SHARE provides information on the amounts of stocks held directly, but not on the amount of stocks through mutual funds and retirement accounts.³⁰ Accordingly, we estimate by maximum likelihood a selectivity-adjusted model for the financial asset share invested in stocks directly, and we find no effect for any of the three cognitive variables at conventional levels of significance.³¹ Therefore, our results imply that cognitive abilities affect the decision to own stocks but not the asset share invested in them.

6. Conclusions

SHARE documents that there is substantial heterogeneity in stock market participation, both within and across countries. As recent evidence for EU countries and the US shows, this is not a pattern particular to SHARE; heterogeneity in stockownership is a widespread phenomenon, with large differences across countries. It bears repeating, however, that SHARE data are collected on a comparable basis for 11 European countries; therefore, they constitute an extremely rich source of information that can be used to investigate the heterogeneity of investment behavior.

In particular, SHARE allows us to examine the relation between cognitive abilities, as measured in the psychological and epidemiological literature, and the decision to invest in stocks, either directly or through mutual funds and other investment accounts. Cognitive abilities might reduce the perceived cost of investing in the stock market, and be correlated with risk tolerance and the ability to process financial data, thus raising stock market participation.

We focus on three domains of cognitive abilities (numeracy, verbal fluency, and memory), and find that they all affect stockownership, even when education is taken into account. SHARE also contains a lot of information related to the life of the elderly in Europe, ranging from physical health, socioeconomic status, intensity of social interaction, and intention to leave bequests. This information gives us the opportunity to quantify the contribution of many different factors in explaining heterogeneity in stockownership in Europe. Even controlling for these and other factors, we find that cognitive impairments reduce the propensity to hold stocks. The effects are statistically significant and economically important.

To explore further the reasons behind the association between cognitive skills and stock market participation, we focus on the effect of cognition on the propensity to hold less information-intensive assets, such as bonds. The results reveal a

²⁹ The Chow test results in the rejection of the hypothesis of equality of coefficients across the two age groups (column 5 in Table A2).

³⁰ For mutual funds and retirement accounts respondents are asked only whether they invest mostly in stocks, about equally in stocks and bonds or mostly in bonds.

³¹ On the other hand, we find that the share of directly held stocks is positively associated with post-secondary education, real wealth and being in a couple. Due to space considerations, these results are available upon request.

weaker association of cognitive abilities with bondholding, which suggests that the association between cognitive abilities and stockholding is likely to be driven by informational constraints, rather than other reasons.

We also report other interesting results. Health status is associated with lower stockownership, while social activities and the intention to leave an inheritance are positively correlated with the decision to own stocks. Finally, the effect of cognitive abilities on participation is larger among the more socially connected individuals, suggesting that social learning and cognitive skills are complementary in enhancing stock market participation.

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Appendix A

A.1. Data construction

We first give a brief description of the financial assets data construction, while full details are given in [Christelis et al. \(2005\)](#). The questions asked on financial assets are as follows: first the respondent is asked whether she owns the asset and then if yes, in what amount. If the respondent refuses to answer the amount question or claims she does not know, then she is introduced into an unfolding brackets sequence which uses three threshold values which differ by country and asset item. The respondent is randomly assigned to one of the three thresholds and she is asked if she owns more or less than that threshold. Depending on her answer, she is then asked about the next higher or lower threshold and so on. The thresholds impose barriers on the range of acceptable values for each asset, which are taken into account during the imputation process ([Table A1](#)).

The imputation procedure involves the construction of a system of equations that include economic and demographic variables, in which each variable is imputed sequentially in many iterations, conditional on the values of the other variables in the system from the same or the previous iterations (for a fuller description of the process see [Christelis, 2008](#)). This chained imputation procedure is analogous to the one implemented in the US Survey of Consumer Finances ([Kennickell, 1991](#)). While numeracy is part of the aforementioned system, fluency, recall, social activities and the probability of leaving an inheritance are not. We impute these variables by regressing them on the full set of demographic and economic variables that have been part of the SHARE imputation process, and generate five alternative imputed values for each missing observation, in order match the five implicate datasets in SHARE. For all three cognitive variables, if the answer to the question is “Don’t know” then there is no imputation performed and we assume that the respondent is not in a position to give an answer to the question (i.e. we assume a wrong answer for numeracy and assign a zero score to fluency and recall).

All values are adjusted for differences in the purchasing power of money across countries using purchasing power parity data from the OECD. We adopt two definitions of stockownership: stocks held directly, or stocks held directly plus stocks held through mutual funds and investment accounts (assuming that holders of mutual funds and individual retirement accounts hold at least some stocks in them). Since the statistical analysis is performed at the household level we choose to combine the information of the two partners in the couple as follows: for the cognitive variables, education, probability of leaving an inheritance, and social activities we take the maximum of the two partners, for health status we take the worse, while for age the average value. The household is denoted as a non-working one if both partners are not working due to retirement, unemployment or otherwise non-participation in the labor force.

A.2. Cognitive abilities

A.2.1. Numeracy

The (abridged) questions on numeracy are as follows. Possible answers are shown in a card while the interviewer is instructed not to read them out to the respondent:

1. *If the chance of getting a disease is 10 percent, how many people out of one thousand would be expected to get the disease?*
The possible answers are 100, 10, 90, 900, and another answer.
2. *In a sale, a shop is selling all items at half price. Before the sale a sofa costs 300 euro. How much will it cost in the sale?* The possible answers are 150, 600, and another answer.

Table A1

Probit regressions for stockholding.

Variable	Direct participation		Total participation	
	Coefficient	Standard error	Coefficient	Standard error
Age	1.649	1.839	2.578	1.865
Age squared	−1.066	1.386	−2.554	1.404
Religious participation	−0.045	0.037	−0.038	0.037
Numeracy	0.094	0.019 **	0.075	0.018 **
Fluency	0.006	0.002 *	0.009	0.002 **
Recall	0.034	0.009 **	0.028	0.009 **
Self-reported health fair or poor	−0.125	0.035 **	−0.068	0.034 *
Depressed last month	0.010	0.030	0.019	0.031
Household income (in 1000 euro)	0.056	0.016 **	0.066	0.016 **
Real assets (in 1000 euro)	0.067	0.009 **	0.048	0.009 **
Self-employed	0.006	0.048	−0.026	0.051
Not working	−0.023	0.044	−0.124	0.043 **
Probability of leaving an inheritance greater than 50,000 euro	0.282	0.048 **	0.271	0.045 **
Couple	−0.068	0.041	−0.164	0.040 **
Social activities	0.068	0.026 *	0.053	0.026 *
<i>Completed some post-secondary education</i>				
Total effect—Sweden	0.128	0.076	0.034	0.108
Incremental effect—Denmark	−0.129	0.127	0.143	0.158
Incremental effect—Germany	0.279	0.119 *	0.175	0.139
Incremental effect—Netherlands	0.106	0.117	0.362	0.140 *
Incremental effect—Belgium	0.091	0.109	0.382	0.133 **
Incremental effect – France	0.260	0.119 *	−0.039	0.148
Incremental effect—Switzerland	0.148	0.154	0.292	0.177
Incremental effect—Austria	0.364	0.172 *	0.445	0.175 *
Incremental effect—Spain	0.184	0.190	0.471	0.190 *
Incremental effect—Italy	0.525	0.172 **	0.420	0.174 *
Incremental effect—Greece	0.201	0.148	0.024	0.163
<i>Gross financial assets (in 1000 euro)</i>				
Total effect—Sweden	0.243	0.025 **	0.485	0.034 **
Incremental effect—Denmark	−0.060	0.035	−0.133	0.049 **
Incremental effect—Germany	−0.014	0.039	−0.161	0.044 **
Incremental effect—the Netherlands	−0.081	0.035 *	−0.275	0.042 **
Incremental effect—Belgium	−0.026	0.032	−0.143	0.040 **
Incremental effect—France	−0.078	0.035 *	−0.078	0.043
Incremental effect—Switzerland	−0.028	0.046	−0.224	0.051 **
Incremental effect—Austria	0.010	0.054	−0.183	0.053 **
Incremental effect—Italy	−0.061	0.046	−0.104	0.057
Incremental effect—Spain	−0.101	0.037 *	−0.218	0.042 **
Incremental effect—Greece	0.014	0.042	−0.138	0.047 *
<i>Country dummies</i>				
Denmark	0.001	0.138	−0.334	0.149 *
Germany	−1.072	0.170 **	−1.269	0.156 **
Netherlands	−0.517	0.142 **	−0.938	0.146 **
Belgium	−0.787	0.133 **	−0.991	0.131 **
France	−0.603	0.142 **	−0.742	0.133 **
Switzerland	−0.603	0.213 **	−0.952	0.209 **
Austria	−1.657	0.231 **	−1.935	0.204 **
Italy	−1.133	0.179 **	−1.856	0.218 **
Spain	−1.048	0.162 **	−1.228	0.143 **
Greece	−1.231	0.169 **	−1.473	0.158 **
Constant (Sweden)	−3.059	0.603 **	−2.368	0.611 **
Observations	19,548		19,548	

Note: The results are a combination of those of five separate regressions (one for each implicate dataset), using the multiple imputation methodology of Rubin (1987). A single star means significance at the 5 percent confidence level while two stars, at the 1 percent level. The R^2 measure is computed as in Estrella (1998).

3. A second-hand car dealer is selling a car for 6000 euro. This is two-thirds of what it costs new. How much did the car cost new? The possible answers are 9000, 4000, 8000, 12,000, 18,000, and another answer.
4. Let us say you have 2000 euro in a saving account. The account earns 10 percent interest each year. How much would you have in the account at the end 2 years? The possible answers are 2420, 2020, 2040, 2100, 2200, 2400, and another answer.

If a person answers (1) correctly she is then asked (3) and if she answers correctly again she is asked (4). Answering (1) correctly results in a score of 3, answering (3) correctly but not (4) results in a score of 4 while answering (4) correctly results in a score of 5. On the other hand if she answers (1) incorrectly she is directed to (2). If she answers (2) correctly she gets a score of 2 while if she answers (2) incorrectly she gets a score of 1 (Table A2).

A.2.2. Fluency

The indicator of fluency is based on the following question: “I would like you to name as many different animals as you can think of. You have one minute to do this.” The indicator is then the number of valid animals named by the respondent. Any member of the animal kingdom, real or mythical is considered a valid answer. Repetitions and proper nouns are instead invalid.

A.2.3. Recall

To measure memory, the interviewer reads a list of 10 items and after a while the respondent is asked which one he or she remembers. The list includes the following items: Butter, Arm, Letter, Queen, Ticket, Grass, Corner, Stone, Book, and Stick. The recall item is the number of items recalled by the respondent.

A.3. Social activities

The SHARE questions on which we base our indicator of social activity is as follows: *Have you done any of these activities in the last month?* Possible answers are: (1) voluntary or charity work; (2) care for a sick or disabled adult; (3) help for family, friends, or neighbors; (4) attendance of an educational or training course; (5) participation in a sport, social, or other kind of club; (6) taking part in a religious organization (church, synagogue, mosque, etc.); (7) taking part in a political or community-related organization; and (8) none of these. The indicator of social activities is the sum of (5) and (7) at the household level (Table A3).

Table A2
Chow tests.

	Direct participation		Total participation		Bondholding
	By age (1)	By sociability (2)	By age (3)	By sociability (4)	By age (5)
Implicate 1	61.6	55.3	119.3 **	69.6 *	93.7 **
Implicate 2	71.6 *	42.2	129.1 **	52.3	91.5 **
Implicate 3	78.4 **	52.5	126.6 **	56.1	88.0 **
Implicate 4	72.2 *	45.8	134.3 **	51.4	86.1 **
Implicate 5	88.2 **	52.9	124.7 **	45.5	83.1 **

Note: The Chow tests were conducted using the procedure suggested by Greene (2002, p. 681). To the best of our knowledge, there is no Chow test that accounts for multiple imputation, and thus we report results from each implicate dataset separately. A single star means significance at the 5 percent confidence level while two stars, at the 1 percent level.

Table A3
Bayesian information criterion for various specifications of the three cognitive variables.

Implicates	Direct participation			Total participation		
	All cognition variables linear	All cognition variables in dummy form	Cognition in dummy form, fluency and recall linear	All cognition variables linear	All cognition variables in dummy form	Cognition in dummy form, fluency and recall linear
Implicate 1	−179,919.6	−179,887.8	−179,899.1	−178,136.5	−178,103.4	−178,108.8
Implicate 2	−180,023.1	−179,985.8	−179,998.8	−178,229.0	−178,196.2	−178,205.7
Implicate 3	−179,877.0	−179,837.0	−179,853.1	−177,981.7	−177,945.8	−177,957.5
Implicate 4	−179,867.8	−179,831.0	−179,843.2	−178,081.2	−178,041.2	−178,055.4
Implicate 5	−179,941.3	−179,905.5	−179,917.8	−178,210.6	−178,174.6	−178,183.2

Note: The criterion is computed as $-2 \ln L - df \ln N$, where L is the likelihood, df the degrees of freedom of the model (equal to the sample size minus the number of parameters) and N the sample size. Smaller values indicate a better fit. To the best of our knowledge, there is no version of the criterion that accounts for multiple imputation, and thus we report results from each implicate dataset separately.

A.4. Probability of leaving an inheritance

The questions that refer to the probability of leaving an inheritance are as follows (the interviewer is instructed to make sure that the respondent includes property and other valuables):

1. *Including property and other valuables, what are the chances that you or your husband/wife/partner will leave an inheritance totaling 50,000 euro or more?* The possible answers range from 0 to 100.
2. *Including property and other valuables, what are the chances that you or your husband/wife/partner will leave any inheritance?* The possible answers range from 0 to 100.
3. *Including property and other valuables, what are the chances that you or your husband/wife/partner will leave an inheritance totaling 150,000 euro or more?* The possible answers range from 0 to 100.

If the respondent gives a zero answer to (1) she is then asked (2), while only if she gives a positive value as an answer to (1) is she asked (3). Our measure of inheritance is the maximum of the chances of leaving an inheritance greater than 50,000 euro across the two partners in a household headed by a couple or the value reported by the household head in a household not headed by a couple.

References

- Ameriks, J., Zeldes, S.P., 2004. How do Portfolio Shares Vary with Age? Mimeo. Columbia University.
- Attanasio, O.P., Banks, J., Tanner, S., 2002. Asset holding and consumption volatility. *Journal of Political Economy* 110, 771–792.
- Barber, B.M., Odean, T., 2001. Boys will be boys: gender, overconfidence, and common stock investment. *Quarterly Journal of Economics* 116, 261–292.
- Barberis, N., Ming, H., Thaler, R.H., 2006. Individual preferences, monetary gambles and stock market participation. *American Economic Review* 96, 1069–1090.
- Benartzi, S., Thaler, R.H., 1995. Myopic loss aversion and the equity premium puzzle. *Quarterly Journal of Economics* 110, 75–92.
- Benjamin, D.J., Brown, S.A., Shapiro, J.M., 2006. Who is Behavioral? Cognitive Ability and Anomalous Preferences. Mimeo. Harvard University.
- Bernheim, D., Garrett, D.M., 2003. The effects of financial education in the workplace: evidence from a survey of households. *Journal of Public Economics* 87, 1487–1519.
- Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegrist, J., Weber, G., 2005. Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Campbell, J.Y., 2006. Household Finance. *Journal of Finance* 61, 1553–1604.
- Christelis, D., 2008. Item non-response in SHARE Wave 2. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Kapteyn, A., Mackenbach, J., Siegrist, J., Weber, G. (Eds.), *First Results from the Survey on Health, Aging and Retirement in Europe (2004–2007): Starting the Longitudinal Dimension*. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Christelis, D., Jappelli, T., Padula, M., 2005. Wealth and portfolio composition. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegrist, J., Weber, G. (Eds.), *Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe*. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Christelis, D., Jappelli, T., Padula, M., 2006. Cognitive abilities and portfolio choice. CSEF Working Paper No. 157.
- Cocco, J.F., Gomes, F.J., Maenhout, P.J., 2005. Consumption and portfolio choice over the life-cycle. *Review of Financial Studies* 18, 490–533.
- Currie, J., Madrian, B., 1999. Health, health insurance and the labor market. In: Ashenfelter, O., Richard, P., Layard, G. (Eds.), *Handbook of Labor Economics*. North-Holland, Amsterdam.
- Dewey, M.E., Prince, M.J., 2005. Cognitive function. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegrist, J., Weber, G. (Eds.), *Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe*. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Dohmen, T., Falk, A., Huffman, D., Sunde, U., 2007. Are risk aversion and impatience related to cognitive ability? IZA Discussion Paper no. 2735.
- Dufo, E., Saez, E., 2002. Participation and investment decisions in a retirement plan: the influence of colleagues' choices. *Journal of Public Economics* 85, 121–148.
- Dynan, K., Maki, D.M., 2001. Does Stock Market Wealth Matter for Consumption? Finance and Economics Discussion Series 2001–23. Board of Governors of the Federal Reserve System, Washington.
- Edwards, R.D., 2008. Health risk and portfolio choice. *Journal of Business and Economic Statistics* 26 (4), 472–485.
- Engstrom, S., Westerberg, A., 2003. Which individuals make active investment decision in the new Swedish pension system? *Journal of Pension Economics and Finance* 2, 1–21.
- Estrella, A., 1998. A new measure of fit for equations with dichotomous dependent variables. *Journal of Business and Economic Statistics* 16, 198–205.
- Frederick, S., 2005. Cognitive reflection and decision making. *Journal of Economic Perspectives* 19, 25–42.
- Goldman, D., Maestas, N., 2005. Medical expenditure and household portfolio choice. NBER Working Paper no. 11818.
- Gollier, C., 2002. What does the classical theory have to say about household portfolios? In: Guiso, L., Jappelli, T., Haliassos, M. (Eds.), *Household Portfolios*. MIT Press, Cambridge, MA.
- Graham, J.R., Harvey, C.R., Huang, H., 2005. Investor competence, trading frequency, and home bias. NBER Working Paper no. 11426.
- Greene, W.H., 2002. *Econometrics Analysis*, fifth ed. Prentice-Hall, Upper Saddle River.
- Guiso, L., Sapienza, P., Zingales, L., 2004. The role of social capital in financial development. *American Economic Review* 94, 526–556.
- Güvenen, F., 2006. Reconciling conflicting evidence on the elasticity of intertemporal substitution: a macroeconomic perspective. *Journal of Monetary Economics* 53, 1451–1472.
- Hank, K., Erlinghagen, M., 2005. Volunteer work. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegrist, J., Weber, G. (Eds.), *Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe*. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Haliassos, M., Bertaut, C., 1995. Why do so few hold stocks? *Economic Journal* 105, 1110–1129.
- Hanushek, E.A., Woessmann, Ludger, 2009. Do better schools lead to more growth? Cognitive skills, economic outcomes, and causation. CESIFO Working Paper no. 2524.
- Hong, H., Kubik, J.D., Stein, J.C., 2004. Social interaction and stock market participation. *Journal of Finance* 59, 137–163.
- Hurd, M., 2002. Portfolio holdings of the elderly. In: Guiso, L., Haliassos, M., Jappelli, T. (Eds.), *Household Portfolios*. MIT Press, Cambridge, MA.
- Jürges, H., 2005a. Cross-country differences in general health. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegrist, J., Weber, G. (Eds.), *Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe*. Mannheim Research Institute for the Economics of Aging, Mannheim.

- Jürges, H., 2005b. Gifts, inheritances and bequest expectations. In: Börsch-Supan, A., Brügiavini, A., Jürges, H., Mackenbach, J., Siegriest, J., Weber, G. (Eds.), *Health, Aging and Retirement in Europe: First Results from the Survey of Health, Aging and Retirement in Europe*. Mannheim Research Institute for the Economics of Aging, Mannheim.
- Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. *Econometrica* 47, 263–291.
- Kaplan, G.A., Turrell, G., Lynch, J.W., Everson, S.A., Helkala, E., Salonen, J.T., 2001. Childhood socioeconomic position and cognitive function in adulthood. *International Journal of Epidemiology* 30, 256–263.
- Kennickell, A., 1991. Imputation of the 1989 Survey of Consumer Finances: stochastic relaxation and multiple imputation. In: *Proceedings of the Section on Survey Research Methods*. American Statistical Association.
- Korniotis, G., Kumar, A., 2006. Does investment skills decline due to cognitive ageing or improve with experience? Mimeo. University of Nôtre-Dame.
- Laibson, D., 1997. Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics* 112, 443–477.
- Li, K.H., Raghunathan, T.E., Rubin, D.B., 1991. Large sample significance levels from multiply imputed data using moment-based statistics and an F reference distribution. *Journal of the American Statistical Association* 86, 1065–1073.
- Lusardi, A., Mitchell, O.S., 2006. Financial literacy and planning: implications for retirement well-being. Pension Research Council WP 2006-1, The Wharton School, University of Pennsylvania.
- Mankiw, N.G., Zeldes, S.P., 1991. The consumption of stockholders and nonstockholders. *Journal of Financial Economics* 29, 97–112.
- Merton, R.C., 1969. Lifetime portfolio selection under uncertainty: the continuous time case. *Review of Economics and Statistics* 51, 247–257.
- Qiu, J., 2004. Precautionary saving and health insurance: a portfolio choice perspective. Mimeo. Wilfrid Laurier University.
- Richards, M., Shipley, B., Furher, R., Wadsworth, M.E., 2004. Cognitive ability in childhood and cognitive decline in mid-life: longitudinal birth cohort study. *British Medical Journal*, 328–552.
- Rosen, H.S., Wu, S., 2004. Portfolio choice and health status. *Journal of Financial Economics* 72, 457–484.
- Rubin, D.B., 1987. *Multiple Imputation for Non-response in Surveys*. Wiley, New York.
- Schae, K.W., 1989. The hazards of cognitive aging. *Gerontologist* 29, 484–493.
- Spaniol, J., Bayen, U.J., 2005. Ageing and conditional probability judgments: a global matching approach. *Psychology and Ageing* 20, 165–181.
- Sims, C., 2003. Implications of rational inattention. *Journal of Monetary Economics* 50, 665–690.
- Smith, J.P., 1999. Healthy bodies and thick wallets: the dual relation between health and economic status. *Journal of Economic Perspectives* 13, 145–166.
- Train, K.E., 2003. *Discrete Choice Methods with Simulation*. Cambridge University Press, Cambridge.
- Van Rooij, Marten, Alessie, Rob, Lusardi, Annamaria, 2007. Financial Literacy and Stock Market Participation. Mimeo.
- Vissing-Jørgensen, A., 2004. Perspectives on behavioral finance: does irrationality disappear with wealth? Evidence from expectations and actions. In: Gertler, M., Rogoff, K. (Eds.), *NBER Macroeconomics Annual 2003*. MIT Press, Cambridge, MA.
- Yates, F.J., 1990. *Judgment and Decision Making*. Prentice-Hall, Englewood Cliffs, NJ.